



The Psychiatric Genomics Consortium: discoveries and directions

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Lancet Psychiatry 2025;
12: 600–10

Published Online
June 26, 2025

[https://doi.org/10.1016/S2215-0366\(25\)00124-5](https://doi.org/10.1016/S2215-0366(25)00124-5)

This online publication has been corrected. The corrected version first appeared at thelancet.com/psychiatry on July 1, 2025

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Research by the Psychiatric Genomics Consortium (PGC) has advanced the discovery of common and rare genetic variations that contribute to the susceptibility to many psychiatric disorders and neurodevelopmental conditions. This Review reflects on major findings from the past 5 years of research by the PGC in five priority areas: discovery of common variants using genome-wide association studies; rare variation and its interplay with polygenic risk; using genetics to go beyond diagnostic boundaries; ascribing functional attributes to genomic discoveries; and developing and implementing processes for data sharing, outreach to various communities, and training. The insights gained in these domains frame the agenda for the next phase of PGC research. In addition to accelerating integrative findings of common and rare variants within, and across, multiple psychiatric disorders and neurodevelopmental conditions, the next phase will use multiple populations to elucidate genetic causes, integrate results with rapidly accumulating multimodal functional genomics data to gain mechanistic understanding, convert genetic findings to clinically actionable phenotypes, such as treatment response, and address the emerging use of polygenic scores. Together, these next steps will highlight the biological underpinnings of psychiatric disorders and neurodevelopmental conditions, which continue to contribute to global morbidity and mortality.

Introduction

The Psychiatric Genomics Consortium

Psychiatric disorders and neurodevelopmental conditions are associated with disability worldwide.¹ Our work builds on decades of family and twin research documenting familial transmission and heritability of psychiatric disorders and neurodevelopmental conditions. The core work of the Psychiatric Genomics Consortium (PGC)—genome-wide association studies (GWAS)—accelerates our understanding of the genetic architecture of psychiatric disorders and neurodevelopmental conditions and yields considerable opportunities to identify mechanisms, develop more personalised interventions, and reduce stigma.²

Comprising over 1770 researchers across 66 countries, the PGC uses coordinated meta-analyses to understand the genetic aetiology of psychiatric disorders and neurodevelopmental conditions. The PGC was established in 2007, has grown considerably in terms of disorders and conditions studied (appendix p 2), and has been enormously productive.³ The appendix (p 7) shows the expanding impact of PGC science. Our 2014 paper on schizophrenia⁴ is among the top ten most cited papers funded by the National Institute of Mental Health.

The fourth phase of PGC science (PGC4) began in 2020, with five objectives (appendix p 8). Encouraged by initial successes, the first objective of PGC4 was to accelerate the discovery of common genetic variants influencing psychiatric disorders and neurodevelopmental conditions in many groups living with such disorders and conditions. PGC4 also investigated the

extent to which copy number variants (CNVs) and other rare and ultrarare variants contribute to the polygenic bases of psychiatric disorders and neurodevelopmental conditions. Although the primary genetic discoveries within the PGC focus on disorders, among the key contributions of PGC4 were insights into the shared genetic aetiology underlying multiple psychiatric disorders and neurodevelopmental conditions. The proposed studies included characterising the genetic aetiology of multiple psychiatric disorders and neurodevelopmental conditions, dissecting heterogeneity within psychiatric disorders and neurodevelopmental conditions, and examining non-diagnostic and cross-diagnostic constructs of clinical relevance (eg, treatment non-response or suicide). To enhance the actionability of our findings, PGC4 integrated omics data generated by others in the field to prioritise causal genes and used causal modelling (eg, Mendelian randomisation) and polygenic scores (indices of genetic risk for an individual relative to the population) to enhance the broader understanding of psychiatric disorders and neurodevelopmental conditions in human health. Lastly, a key objective in PGC4 was to widen the reach of these scientific discoveries by accelerating data sharing, training investigators from around the world to conduct GWAS, and sharing information about the real-world impact of psychiatric genetics to the broader community (appendix p 9). This Review outlines some of the key insights that the PGC has accumulated across these five objectives, shows how they inform emerging priority areas in psychiatric genetics, and presents the PGC's scientific agenda.

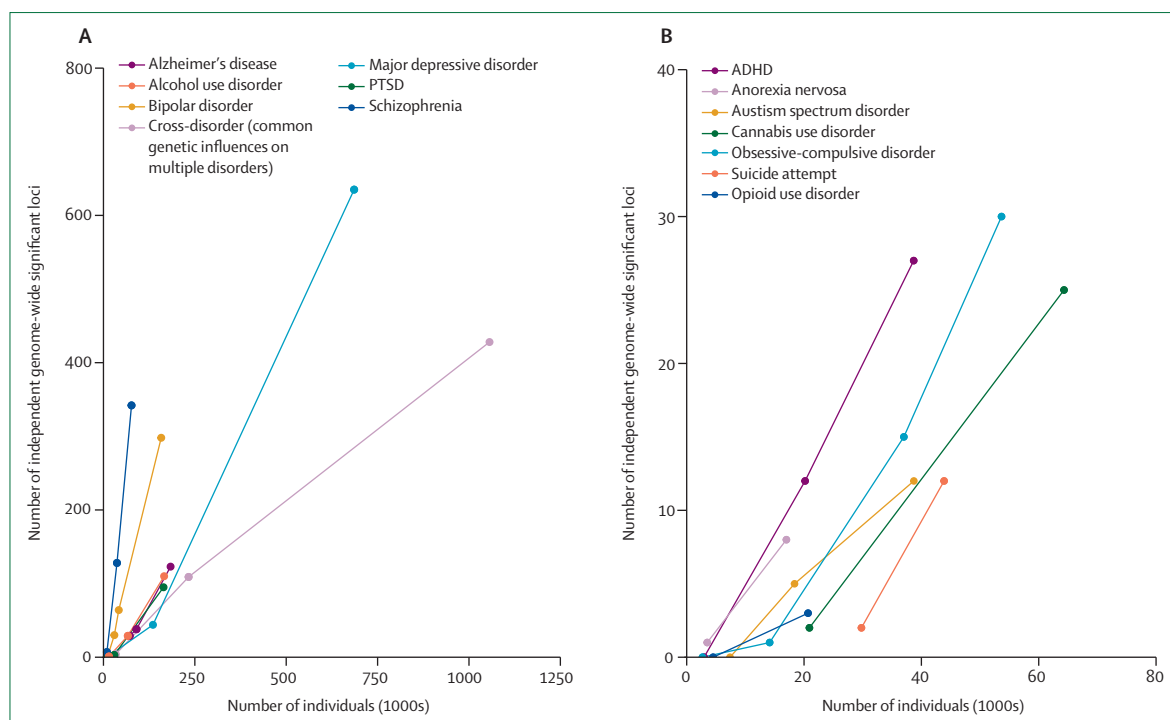


Figure 1: Identification of genome-wide significant loci with time and increasing sample size

(A) GWAS that have 90 or more genome-wide significant findings. (B) GWAS with fewer than 90 genome-wide significant findings ($p \leq 10^{-5}$). For both panels, each datapoint is a published GWAS. In all cases, increases in sample sizes have yielded approximately linear increases in loci discovery. Raw estimates for each datapoint can be found in the appendix (p 10). Disorders with only one published GWAS were not included (ie, anxiety disorder, eating disorder [specifically binge-eating], and Tourette's syndrome). GWAS=genome-wide association studies.

Common genetic variation: discoveries

PGC4 has advanced genetic discoveries across a range of psychiatric disorders and neurodevelopmental conditions.³ During the past 5 years, the number of loci identified has increased substantially (figure 1), reflecting increasing numbers of cases, exceeding 100 000 for many disorders and conditions (appendix p 10). Each iteration of analysis added individuals whose genomes represent a wider collection of global communities affected by these psychiatric disorders and neurodevelopmental conditions (figure 2). The number of loci per effective sample size varied by disorder—common disorders of moderate heritability (eg, major depressive disorder) require larger sample sizes than less common disorders of higher heritability (eg, schizophrenia) to identify the same number of risk loci. The latest GWAS of schizophrenia included 76755 individuals with schizophrenia or schizoaffective disorder, 243 649 control individuals, and identified 287 distinct genome-wide significant loci.⁵ Major depressive disorder, which did not progress as quickly as other psychiatric disorders in locus discovery, has now been associated with 635 genome-wide significant loci (688 808 individuals across multiple genetic ancestries with depression and over 4 million control individuals).⁶ For schizophrenia, bipolar disorder, and major depressive disorder, each successive iteration has largely validated signals found in previous

GWAS—a finding that indicates stability alongside discovery. By contrast, GWAS of eating disorders, obsessive-compulsive disorder, and Tourette's syndrome, although successful at identifying loci, are still in early phases of discovery (figure 1; appendix p 10).

Beyond increasing the number of loci, each subsequent PGC GWAS has provided aetiological insights (appendix p 2) detailing potential biological mechanisms. Importantly, these GWAS have pointed towards systems that can potentially be altered by pharmacotherapeutics, thereby establishing a framework for testing the impact of GWAS on the drug discovery pipeline, bringing us a step closer to action and translation.

Common genetic variation discovery: next steps

Despite the success of PGC GWAS for many psychiatric disorders and neurodevelopmental conditions, no GWAS has reached the point of diminishing returns or saturation.⁷ For several disorders and conditions, particularly those studied in the past 5–10 years (eg, eating disorders, obsessive-compulsive disorder, and Tourette's syndrome), GWAS have not reached a point of discovery where future novel loci are unlikely to yield additional biological insights (figure 1), therefore increasing sample sizes remains the highest priority. Estimates of the number of independent

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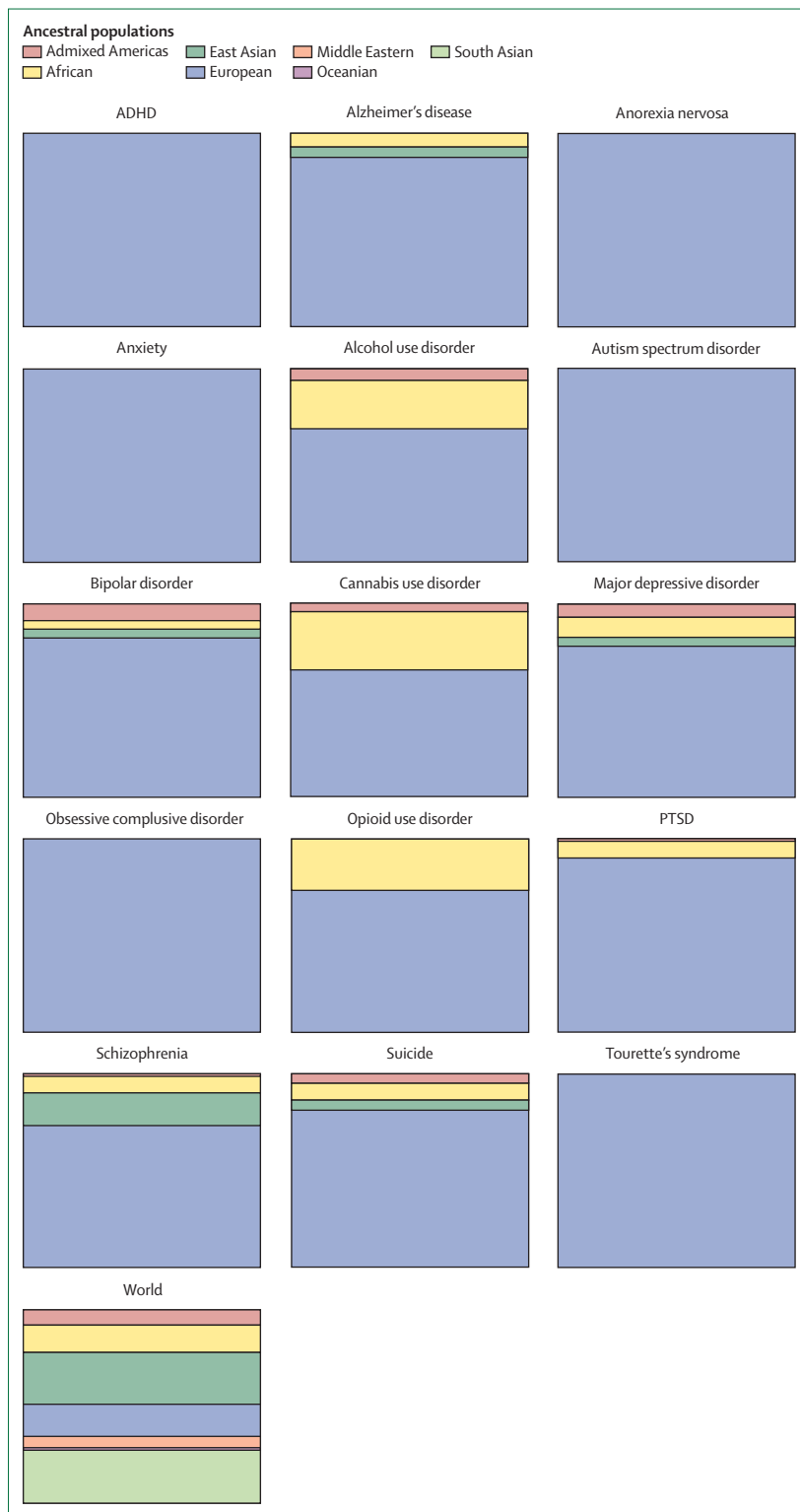


Figure 2: Global data from ancestral populations in Psychiatric Genomics Consortium publications
Correct as of Dec 20, 2024.

common risk loci contributing to disorders such as schizophrenia, major depressive disorder, and ADHD,^{8,9} suggest that, even given success to date of the relevant GWAS, more loci remain to be identified before the breadth of their common variant genetic aetiologies is fully integrated.

GWAS of avoidant–restrictive food intake disorder,¹⁰ substance use disorders (eg, stimulant and cocaine use disorders), postpartum depression,¹¹ and childhood versus adult presentations of disorders and conditions¹² are underway. GWAS of quantitative symptom measures, age at diagnosis,^{13,14} and specific symptoms constituting a diagnosis,¹⁵ and sex-specific GWAS are essential to dissect the heterogeneity within, and across, disorders. Therefore, new data collection to increase sample size should include the detailed phenotyping needed to investigate the biological bases of disorder subtypes.

Expanding the global impact of GWAS discoveries remains a key scientific goal. Although substantial progress has been made in this domain over the past 5 years (figure 2), the number of loci discovered in populations worldwide is far behind findings in those of recent European ancestry. There are many scientific advantages to studying multiple ancestral populations, including the existence of novel variants, differences in the allele frequency spectrum, and linkage disequilibrium structure. These genetic features aid in identification of the actual causal variants within loci (ie, fine mapping) and of environmental influences.^{16,17} A global approach is crucial to optimise polygenic scores: incorporating GWAS of different ancestries in the development and estimation of polygenic scores has been shown to improve their performance in independent test cohorts across ancestries.^{18,19} For example, using both European and East Asian GWAS of schizophrenia to train polygenic scores yielded a median R^2 variance explained of 6·2% in East Asian test cohorts versus 3·9% when polygenic scores were developed using East Asian GWAS alone.¹⁹ These scores, when combined with multimodal clinical data and information on other types of genetic variation, such as CNV, could aid personalisation of treatments and early interventions—an aspect of polygenic score research that PGC will pursue. The PGC will continue to incorporate data from new biobanks (eg, All of Us research programme)²⁰ and foster collaborations with psychiatric genetics initiatives across the world (eg, Ancestral Populations Network)²¹ to achieve these scientific objectives.

Discoveries with rare variants in the context of common variants

Consistent with polygenic models of psychiatric disorders, a majority of PGC effort has concentrated on identifying and characterising common genetic variation. The impact of uncommon (allele frequency <0·01) and rare or ultrarare (<0·001) variants, and the extent to which they aggregate with and augment the effects of polygenicity,

was a core focus of PGC4.^{22,23} Most PGC discovery efforts focused on CNVs with the development and implementation of a pipeline for identification and analysis of CNVs from GWAS datasets. The PGC documented an overall enrichment of rare CNV in people with schizophrenia compared with control CNVs. The gene sets that were identified, particularly those involving deletions, were enriched for genes encoding neuronal and synaptic activity.²⁴ In a follow-up study, the PGC identified sporadically occurring somatic CNV in exons 1–5 of *NRXN1*, a gene associated with autism spectrum disorder and intellectual disability, and a recurrent intragenic deletion in *ABCB11*, which is involved in response to antipsychotics.²⁵

CNVs have proven to be important in multiple commonly occurring disorders. For instance, the genome-wide neurodevelopmental CNV burden explained an additional 0.34% of phenotypic variance in post-traumatic stress disorder (PTSD) beyond the polygenic signal, with notable associations with a del(15)(q11.2) BP1–2 microdeletion.²⁶ CNVs in individuals with anorexia nervosa are enriched in genes involved in synaptic function, metabolic and mitochondrial factors, and lipid characteristics, consistent with the metabopsychiatric conceptualisation of the disorder.²⁷ Supporting the importance of CNVs across a range of psychiatric conditions, genome-wide analyses for autism spectrum disorder, ADHD, bipolar disorder, major depressive disorder, PTSD, and schizophrenia identified 36 genome-wide significant associations (Sebat J, unpublished). Some CNV alleles had opposing associations (del[15](q11.2) with risk for schizophrenia, but protective for bipolar disorder), whereas others conferred risk to more than one disorder or condition (dup[15](q11.2) for major depressive disorder and autism spectrum disorder). A comparative analysis of rare CNV burden across these disorders and conditions suggest some specificity. Schizophrenia, in particular, showed distinct pathway and tissue associations with CNVs. Duplications were concentrated in developmental signalling pathways, genes expressed in neural precursor cells and frontal cortex, and deletions were concentrated in synaptic function pathways, and genes expressed in maturing neuronal cells and occipital cortex.

Another discovery approach has involved sequencing densely affected pedigrees,^{28–30} including the identification of multiple ultrarare missense variants segregating in schizophrenia pedigrees.²⁹ In an 11-generation Tourette's syndrome pedigree, PGC investigators identified one rare and three ultrarare deleterious variants in genes implicated in neurodevelopmental processes.³⁰ A new resource, PECAN, was developed to integrate CNV calling and analysis in these large, multiplexed pedigrees, enabling greater range of variant detection.³¹

Although not funded to generate sequence data, the PGC collaborated with consortia generating such data, including schizophrenia exome meta-analysis,³² WeSeq (R01 MH130671), Blended Genome Exome,³³ Bipolar

Exome,³⁴ and the Million Veteran Program.³⁵ These PGC analyses have focused on integrating variation across the allelic spectrum. Of interest is whether rare variation occurs in regions previously implicated as genome-wide significant for common variants. For schizophrenia, bipolar disorder, and autism spectrum disorder, moderate signal convergence suggests that significant GWAS loci ($p \leq 10^{-8}$) provided a reasonable guide in the search for rare variation.^{32,36,37} For instance, for schizophrenia, genes with significantly more ultrarare coding variation were also more likely to occur in genes prioritised by bioinformatic analysis for GWAS (eg, *GRIN2A*). The 2025 bipolar disorder GWAS yielded odds ratios of approximately 1.2 (95% CI 1.05–1.28) for common and rare variant convergence.³⁶

The second question of interest is how rare variants and polygenic risk jointly influence psychiatric traits within an individual. For instance, PGC investigators found that risk for autism spectrum disorder was attributable to the combined effects of de novo protein-truncating and missense mutations, rare disruptive inherited variants, and polygenic scores, with combined rare and common variation each explaining about half of the 4% variance in case status in the condition.³⁸ Consistent with a liability threshold model, polygenic scores are inversely correlated with rare variation in both autism spectrum disorder³⁸ and schizophrenia.³⁹

CNV and rare variant analyses: next steps

The recent PGC CNV findings of genetic commonality and specificity across six disorders (Sebat J, unpublished), highlighted the need to expand common variant cross-disorder and cross-condition genetic analyses to the study of CNVs and other forms of rare variation. The CNV working group intends to extend CNV calling and analysis of all the psychiatric conditions and neurodevelopmental conditions within the PGC, using whole-exome sequencing and whole-genome sequencing data, where available, alongside GWAS.

PGC studies supported evidence for genic convergence for different types and frequencies of causal variation within, and between, disorders and conditions.⁴⁰ Novel quantitative approaches (eg, Bayesian models) add robustness to such integrative analyses; however, to have good statistical power to investigate the effects of rare and common variants in combination requires sample sizes larger than those needed for the initial discovery of the variants. Genotype and sequencing data have been collected on all PGC disorders and conditions through large biobanks, such as UK Biobank⁴¹ and All of Us,²⁰ and public research efforts such as the Million Veterans Program.⁴² New initiatives, such as the Genes 2 Mental Health Network⁴³ and Simons Searchlight,⁴⁴ take a so-called gene-first approach to investigating genetic effects on neurocognitive and psychiatric traits. The PGC is collaborating with these new initiatives to conduct well powered studies of combined influences of

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See Online for appendix

CNVs, gene mutations, and polygenic risk on clinical outcome.

Using genetics to go beyond diagnostic boundaries

Substantiating findings from twin and family studies, single nucleotide polymorphism-based genome-wide genetic correlations estimated by the PGC across 13 major psychiatric disorders and neurodevelopmental conditions suggested widespread pleiotropy.⁴⁵ This finding prompted the PGC to consider genetic commonality across disorders and conditions. Several studies were conducted, the results of which challenge categorical diagnostic confines. Patterns of genetic correlations, which were also evident in sex-stratified analyses, indicated a high degree of pleiotropy across many psychiatric disorders and neurodevelopmental conditions.^{5,46–50}

The PGC Cross Disorder Analysis Working Group parsed the genetics of eight disorders into three primary and inter-related genetic domains corresponding to (1) mood and psychotic disorders (major depressive disorder, bipolar disorder, and schizophrenia); (2) compulsive behaviour disorders (anorexia nervosa, obsessive-compulsive disorder, and Tourette's syndrome); and (3) early onset disorders and conditions (autism spectrum disorder, ADHD, Tourette's syndrome, and major depressive disorder).⁴⁵ In subsequent analyses, the group aggregated genomic data on 11 psychiatric disorders and neurodevelopmental conditions.⁵¹ In 2025, the group studied 14 psychiatric disorders and neurodevelopmental conditions with data from more than 1 million individuals (preprint).⁵² Five latent genomic factors captured the majority of genetic variation across the 14 disorders and conditions: (1) compulsive (anorexia nervosa, obsessive-compulsive disorder, Tourette's syndrome, and anxiety); (2) psychotic (schizophrenia and bipolar disorder); (3) neurodevelopmental (autism spectrum disorder, ADHD, and components of Tourette's syndrome); (4) internalising (major depressive disorder, anxiety, and PTSD); and (5) problematic substance involvement (alcohol, tobacco, cannabis, and opioid use disorders, and, to a lesser extent, ADHD).

Combining disorders and conditions to increase statistical power and subsequent detection of broad genetic signals led to the identification of 428 pleiotropic loci. These genetic signals were enriched in general biological processes (eg, transcription regulation) and each of the individual genomic factors enriched in genes expressed in specific brain cell types. These results suggest a hierarchy of shared genetic effects, from a general predisposition to psychiatric disorders and neurodevelopmental conditions to more specific effects on subdomains of psychopathology.

Additional analyses have tested specific hypotheses.³⁶ For example, compulsive exercise in volunteer population cohorts, notably in girls aged 14–16 years, was discovered

to be associated with polygenic scores for anorexia nervosa and obsessive-compulsive disorder.⁵³ Mendelian randomisation analysis suggested bidirectional causal genetic effects and pleiotropy between schizophrenia and cannabis use disorder.⁵⁴ Polygenic scores for PTSD were associated with postpartum depression, even after accounting for genetic risk for major depressive disorder⁵⁵ and genetic risk for ADHD was found to have a potentially causal effect on PTSD, but not vice versa.⁵⁶ Substance use disorder analyses across four substances yielded both cross-cutting addiction loci and substance-specific loci.⁵⁷ Cross-disorder and cross-condition analyses for ADHD and autism spectrum disorder identified both shared and unique variants: in which the shared genomic proportion was strongly correlated with other psychiatric disorders and neurodevelopmental conditions beyond ADHD and autism spectrum disorder, but the differentiating portion was best accounted for by cognitive traits.⁵⁸

Another goal of PGC4 was to study the genetic architecture of transdiagnostic constructs, such as suicidal behaviour—a phenotype with high relevance to public health. Approximately 80% of individuals who attempt suicide have a lifetime psychiatric disorder,⁵⁹ which presents opportunities to understand not only how genetic variation contributes to common behaviours across disorders, but also whether clinically and genetically distinct subtypes of suicidal behaviours are evident across these disorders. In 2023, the PGC Suicide Working Group led the largest GWAS meta-analysis of suicide attempt, establishing the genetic overlap of suicidal behaviours (both attempt and death) with nearly all psychiatric disorders.⁴⁸ Moreover, suicide attempt and death each had common and unique associations with psychiatric disorder phenotypes.

The challenges posed by genetically similar conditions that are diagnostically distinct are compounded by within-disorder heterogeneity (eg, individual symptoms), for which few large-scale data are available. Where these data were available, results have affirmed concerns that diagnostic standardisation has diluted meaningful clinical heterogeneity. For example, genetic factors associated with appetitive, vegetative, cognitive, and gating symptoms were identified for major depressive disorder, with the appetitive genetic factor being specifically correlated with BMI and tobacco smoking (preprint).⁶⁰ Other studies of clinically acknowledged or statistically inferred subtypes (eg, persistent ADHD vs late-diagnosed ADHD;¹² restricting anorexia nervosa vs binge-purge anorexia nervosa⁶¹) have documented support for corresponding genetic heterogeneity. Additional data on severity, chronicity, comorbidity, and ascertainment might aid in refinement of genetic architecture.^{36,62}

GWAS of psychiatric disorders and neurodevelopmental conditions also identify genetic liability that meaningfully influences continuous variation in human behavior. Extreme differences of this distribution likely underpin

multiple psychiatric disorders and neurodevelopmental conditions. For instance, PGC researchers found that polygenic scores for multiple psychiatric disorders and neurodevelopmental conditions were associated with externalising and internalising symptoms in children aged 9–11 years (preprint).^{63,64} Related analyses supported genetic commonality between psychiatric disorders and behavioural traits (eg, major depressive disorder and neuroticism⁶⁵). Other studies have found key differences: for example, alcohol use disorder is genetically correlated with psychopathology, whereas alcohol consumption is more closely related to socioeconomic status.⁶⁶ Polygenic scores can be used as a proxy for unmeasured phenotypes. These scores can be particularly important for neurodevelopmental conditions, such as ADHD and autism spectrum disorder, for which phenotypic measures are often unavailable in adults and cannot be extracted from health registers since the disorders are still largely underdiagnosed in the adult population.^{67,68}

Cross-disorder genetics: next steps

Genetic heterogeneity within a disorder can occlude the identification of core and cross-cutting diagnostic features, which might be ideal treatment targets for difficult to manage comorbid conditions. Just as individual-disorder GWAS have led to clues about potentially new pharmacotherapeutics, the identification of loci that influence multiple disorders, or conversely, are relevant to only certain forms of the same disorder, could profoundly impact drug repurposing efforts and selection of treatments. Looking forward, the PGC anticipates evaluating the extent to which all forms of variation (common, rare, and structural) contribute to this shared or distinct genetic landscape. The PGC plans to build on previous work^{69,70} to delineate genetic influences that shape resilience to psychiatric disorders, in the context of both genetic and environmental risk. Such resilience goes beyond the absence of disorder,⁷¹ to outline the conditions in which individuals thrive despite predisposition and adversity. Sources of heterogeneity remain implicated in the epidemiological studies that the PGC continues to investigate from a genetic standpoint, such as developmental course (eg, early onset and adolescent delimited⁷²) sex differences,^{46,73} severity, recurrence (eg, persistent symptoms *vs* remitting and relapsing), and symptom groups.¹⁵

Integration of GWAS with functional genomics to accelerate discovery and impact

GWAS loci can comprise multiple significant and correlated variants, but few of these variants alter protein coding. Determining the biology underlying GWAS loci requires additional functional omics data, especially from various global populations to aid fine mapping. Our systematic approach to functional characterisation relies on coordinated intake of omics data that are then subjected to a validated quality control and analysis

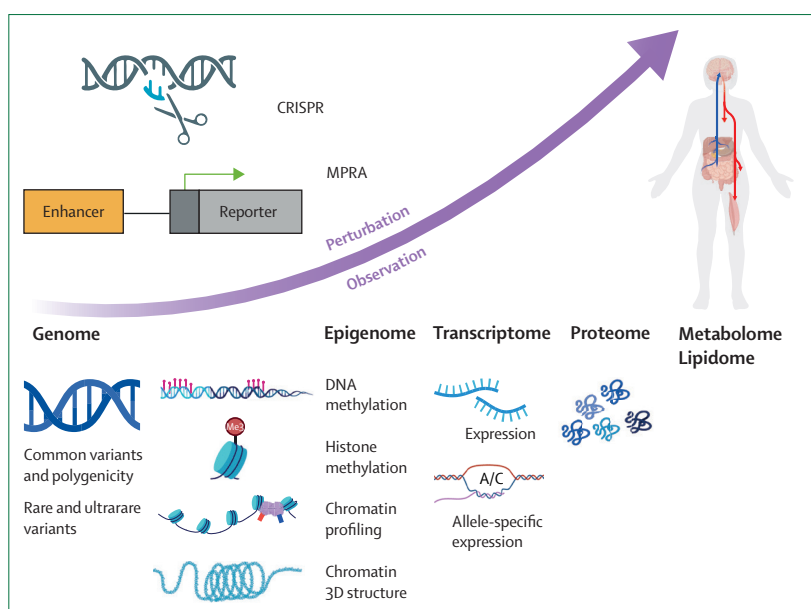


Figure 3: Integration of multimodal functional genomics data with Psychiatric Genomics Consortium GWAS Established and emerging high-throughput technologies enable the generation of assorted datasets across multiple biological layers, including the genome, epigenome, transcriptome, proteome, metabolome, and lipidome. QTL analyses, such as expression QTLs and protein QTLs, link genetic variants to molecular traits, providing insights into the genetic regulation of biological processes. These data can be integrated with GWAS findings and case-control or cohort studies to explore the combined effects of genetic and environmental factors on complex traits. Perturbation-based approaches—such as CRISPR screens or high-throughput reporter assays (eg, MPRA)—allow functional validation of candidate causal genes and variants by testing their effects on molecular phenotypes or cellular functions. Together, these multimodal datasets support a systems-level understanding of psychiatric disorders and neurodevelopmental conditions, and help to identify mechanistic pathways. Figure created with Biorender.com. GWAS=genome-wide association studies. MPRA=massively parallel reporter assays. QTL=quantitative trait loci. 3D=three-dimensional.

pipeline. Such a systematic approach can offer insights into commonalities across, and distinctions between, disorders at the molecular and cellular levels.

Variants in genome-wide loci have been found to cluster in regulatory regions of the genome. Transcriptome-wide association analyses pair disease-variant associations with variant-gene expression associations drawn from large omics datasets (eg, GTEx⁷⁴ and MetaBrain⁷⁵). Such approaches using gene expression quantitative trait loci (eQTL) have been extended to, and integrated with, other types of omics data (figure 3; appendix p 13). The anxiety GWAS combined GWAS results with eQTL, methylation QTL, and protein QTL data to identify 66 genes with high priority for anxiety (preprint).⁷⁶

With the growing availability of cell-type-specific omics data, PGC GWAS have incorporated functional characterisation at cellular resolution, testing for over-representation of risk variants near genes or regulatory elements active in specific cell types to identify the cells likely involved in disease aetiology (enrichment analyses). The latest bipolar disorder GWAS identified enrichment of risk variants in amygdala excitatory and inhibitory projection neurons, hippocampal dentate gyrus neurons, cortical γ -aminobutyric-acid (GABA) interneurons, and in enteroendocrine cells of the large

For the PGC Video Textbook see
<https://pgcanalytics.github.io/pgcvideoebook/>

For the GWAS summary
 statistics see <https://pgc.unc.edu/for-researchers/download-results/>

intestine and delta cells of the pancreas.³⁶ For ADHD, major depressive disorder, and obsessive-compulsive disorder GWAS have shown enrichment of risk variants in both inhibitory and excitatory neurons;^{6,9,77} for PTSD GWAS showed enrichment particularly in mature and developing GABAergic neurons;⁷⁸ for ADHD, the primary cell type indicated was midbrain dopaminergic neurons.⁹ Besides the dominant neuronal contributions to the causes of psychiatric disorders, emerging immunogenetic datasets from blood have offered insights into the role of immune cells in psychiatric disorders, including a transdiagnostic role for activated T cells.⁷⁹

Spatiotemporal expression data have provided developmental insights into the causes of psychiatric disorders. Genome-wide analyses across multiple psychiatric disorders found that pleiotropic risk genes exhibited peak chromatin accessibility in the second trimester, whereas disorder-specific genes—such as those implicated in schizophrenia—had peak chromatin accessibility during the first trimester.⁴⁵

Next steps to expand functional interpretation of GWAS

Anticipating rapid accelerations in the generation of brain functional genomics data (eg, PsychENCODE2),⁸⁰ the PGC formed the functional genomics working group. This group develops and implements a standardised procedure for intake and quality control of omics data to serve three broad purposes: first, as a repository that each future PGC GWAS could use to further annotate their discoveries; second, for the working group to characterise the functional landscape across psychiatric disorders and neurodevelopmental conditions; and third, to make functional genomic resources more widely available. Incoming data types will address various modality-specific questions (figure 3; appendix p 13) and integrative research questions (eg, immune or metabolic pathways from multiple single-cell data frames), thereby focussing on integrating phenotypic data into the working group's analyses.

Generating actionable science in the PGC

A key objective of PGC4 was to ensure that the science conducted by the PGC was accessible to a broad audience; for example, via social media platforms (appendix p 9). The PGC Outreach Committee has worked to improve the visibility, accessibility, and understanding of psychiatric genetics concepts, as well as to remove barriers to participation in psychiatric genetics research.^{81,82} The group has led educational efforts for many audiences, including researchers intending to add genetic findings to their investigations; clinicians interested in incorporating discussions about genetics into their interactions with patients and their families; and the broader community of patients and participants who wish to learn more about genetic contributions to mental health. In support of this objective, the open

access PGC Video Textbook was launched in 2024, accompanied by transcripts in English and Portuguese, and is open to further translation.

The positive impact of data sharing and open science

The PGC is a successful demonstration of team science and its impact extends beyond the large-scale GWAS of psychiatric disorders and neurodevelopmental conditions. Summary statistics for more than 50 GWAS are freely downloadable and have been accessed more than 154 000 times since 2021. This data-sharing policy, has led to the routine inclusion of psychiatric disorders and neurodevelopmental conditions in many genomic, functional genomic, and biological studies. Genes identified through unbiased GWAS are now frequently studied in cross-species behavioural paradigms.^{83–85} Innovative experiments conducted in induced pluripotent stem cells derived from high and low polygenic backgrounds (eg, for both schizophrenia and alcohol use disorder) have provided unprecedented opportunities to characterise the cellular switches that influence systems underlying psychiatric disorders.^{86,87} Ascertainment of signal enrichment in vast amounts of brain-derived single-cell data generated by National Institute of Health initiatives has relied on PGC GWAS.⁸⁸ The PGC currently has 418 datasets of genome-wide significant loci (332 individual-level and 86 summary-level), representing over 400 000 individuals with psychiatric disorders and neurodevelopmental conditions and 1·5 million control individuals. These data have been accessed via 772 analysis proposals from over 250 investigators. All data analysis tools supported by the PGC can be downloaded from open sources, such as GitHub.

Progress towards clinical translation

The widespread feasibility of computing polygenic scores has accelerated their inclusion in various studies investigating clinical outcomes. Schizophrenia polygenic scores, for example, have been linked to clozapine dosage,⁸⁹ ADHD polygenic scores to treatment-resistant depression,⁹⁰ alcohol use disorder polygenic scores to non-abstinent remission,⁹¹ and age at menarche polygenic scores with onset of anorexia nervosa at a young age.¹⁴ Using curated datasets that evaluate gene expression differences in cell lines exposed to perturbagens, the most recent PGC GWAS have included analyses that prioritise druggable genes as pharmacotherapeutic targets.^{6,36,48,57}

The PGC remains focused on translating genetic discoveries into tools for improved clinical care. A promising use case is the selection of antipsychotic drugs in schizophrenia, which is currently based on a trial-and-error approach.⁹² Thus, patients are often treated with antipsychotic drugs that have little therapeutic effect and several side-effects, despite the availability of clozapine—an effective antipsychotic medication that is underused because of rare but serious side-effects that

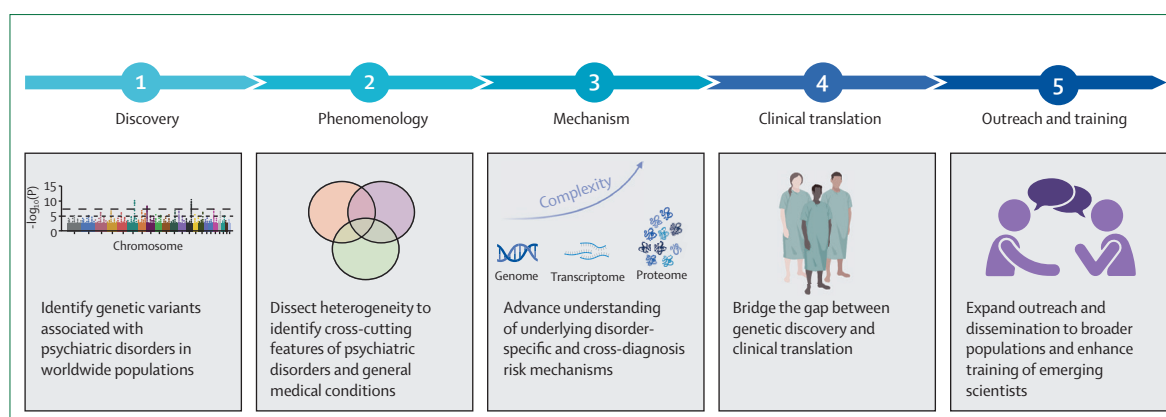


Figure 4: Research agenda for phase 5 of the Psychiatric Genomics Consortium

Insights from PGC4 have directed the research plan for the next phase of the Psychiatric Genomics Consortium. First, we will substantially increase locus discovery with larger samples from global populations to identify novel variants across the allele frequency spectrum and analyse these common and rare variants. Second, we will disarticulate those aspects of multiple psychiatric disorders and neurodevelopmental conditions that are shared from those that contribute to within-disorder and between-disorder heterogeneity. Using large-scale common and rare variant data to characterise phenomenology will allow us to connect psychiatric disorders and neurodevelopmental conditions with other chronic medical conditions (eg, cardiometabolic), with which they often co-occur. Third, we plan to use increasing amounts of functional omics (eg, epigenetic) data to delineate the biological mechanisms underlying psychiatric disorders and neurodevelopmental conditions, and to identify pathways or omics features that influence multiple disorders and conditions. Fourth, we will use aggregate genomic data, such as polygenic scores, to study clinically relevant phenotypes (eg, treatment response). To enable early monitoring for risk, we will use polygenic scores with other environmental data to characterise longitudinal trajectories of risk and resilience. We will continue to inform all relevant scientific and clinical communities of our findings and maintain our strong tradition of psychiatric genetics training. Figure created with Biorender.com.

require intensive monitoring. Several PGC schizophrenia samples include medication and outcome data, presenting the opportunity to develop polygenic and multimodal risk profiles to identify patients for whom early use of clozapine might be justified^{93,94} by predicting treatment-resistant schizophrenia and assessing severe side-effects (eg, agranulocytosis).^{95,96}

Psychiatric disorders and neurodevelopmental conditions affect individuals across the lifespan: the PGC studies disorders and conditions that show onset in childhood, adulthood, and older age. Understanding the aetiology of prognosis of early-onset disorders and conditions is central to facilitating primary and secondary prevention of psychopathology. However, genomic contributors to longitudinal trajectories and prognosis remain incompletely characterised.^{97,98} Building on our network of large, international, and longitudinal samples, we can identify the genomic antecedents of disorder onset and progression by applying transdiagnostic and disorder-specific polygenic methods. When combined with additional multimodal risk predictors, such analyses can help isolate periods of heightened sensitivity when prevention and treatment might have the most effect.⁹⁹ The inclusion of a range of potentially modifiable factors (eg, environmental stressors and lifestyle), which might be causally associated with reduced risk of incident disorder and improve prognosis among affected individuals, could provide further targets for prevention.¹⁰⁰

What's next for the PGC?

The next phase of PGC science will widen the scope of genetic discoveries for multiple psychiatric disorders and

their responses to treatment. The identification of novel common and rare single nucleotide and structural variants is anticipated for every disorder and condition that we study, given the increasing availability of data. By studying these individual disorders and conditions, we will parse genetic components that associate disorders and conditions with one another and those that differentiate aspects of a single disorder or condition into differing genetic aetiologies. As new sources of omics data become available, our individual and cross-disorder findings will increasingly be translated into biological insights. Using both genetic and other sources of environmental, clinical, and neuroscience data, we are optimistic that the PGC's lifespan approach, which includes disorders and conditions that typically onset in childhood (eg, autism spectrum disorder and ADHD), adulthood (eg, schizophrenia, major depressive disorder, and substance use disorder), and older age (eg, Alzheimer's disease), will help us to examine the longitudinal course of psychiatric health and resilience in populations of differing attributes. Our scientific journey from discovery to phenomenology, mechanism, and clinical application is based on broad scientific collaboration and rigorous methodology. We will disseminate our findings across relevant scientific and clinical communities (figure 4). Our work will drive advancements across the field, establish a robust platform for future research, inspire innovation, and promote collaborative, transparent, and reproducible science to enhance clinical impact.

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Contributors

This Review was initially drafted by AA, CMB, and PFS. Content was supplied from all named authors, with input from all coordinating committee members. AA and CMB coordinated all aspects of this Review. All named authors edited the draft. AA and CMB had final responsibility for submission upon approval by all named authors.

Declaration of interests

CMB is a royalty recipient from Pearson Education and is supported by the National Institutes of Mental Health (NIMH; R01MH136149, R01MH134039, R56MH129437, R01MH120170, R01MH124871, and R01MH124871). OAA is a consultant to Cortechs.ai and Precision Health; has received speaker's honoraria from Lilly, Janssen, Lundbeck, and Otsuka; has a patent for Intranasal Administration (US20160310683 A1); and is the National Principal Investigator for clinical trials: esketamine treatment in depression (Janssen Pharmaceuticals), MDMA-assisted therapy for PTSD (MAPS), and Connex trial for cognition in schizophrenia (Boehringer Ingelheim). KWC is a consultant to Sword Health and is an unpaid board member of the International Society of Psychiatric Genetics. HLD reports funding from Novo Nordisk Foundation. LKD reports honoraria from the University of North Carolina, University of Texas, and University of Pennsylvania. BF has received educational speaker fees and travel support from Medice. M-EL receives royalties from Cambridge University Press. MCO'D reports a research grant from Akviva Health and has received a research grant from Takeda Pharmaceuticals within the past 3 years. JWS is a member of the Scientific Advisory Board of Sensorium Therapeutics (with stock options) and has received grant support from Biogen. JTRW consults, and has a research grant from, Akviva Health and has received a research grant from Takeda Pharmaceuticals within the past 3 years. PFS is a consultant and shareholder of Neumora Therapeutics (past 3 years). OAA reports funding from the Research Council of Norway (324252 and 324499), EU Horizon Research and Innovation Action Grants (964874), Stiftelsen Kristian Gerhard Jebsen, NIMH (51R01MH124839), South-East Norway Health Authority (2023-031), Nordforsk (164218), and Nasjonalforeningen. ARD reports funding from NIMH (R01MH123619 and R01MH132733). PG-R reports funding from NIMH (U01MH125062 and R01MH125236) and the Brain and Behavior Research Foundation. JMH reports funding from NIMH (5R01MH124847). JG reports funding from the National Institute of Drug Abuse (NIDA; R01DA058862). JH-L reports funding from the Knut and Alice Wallenberg Foundation (KAW 2023-0288), Swedish Research Council (2024-03567 and 2022-00445), and Swedish Brain Foundation (FO2024-0144-HK-49). ECJ is supported by funding from NIDA (K01DA051759). CML reports funding from NIMH (R01MH124871) and the Wellcome Trust (226770/Z/22/Z). AMM reports funding from the Wellcome Trust (226770/Z/22/Z, 223165/Z/21/Z, and 220857/Z/20/Z) and UK Research and Innovation (MR/Z000548/1, MR/Z50354X/1, MR/Z503563/1, and MR/W014386/1). JLM-O reports funding from NIDA (1DP1DA058737), the US Department of Veterans Affairs (IK2 CX002095), and the National Institutes of Health (NIH; 1OT2OD037644-01). NM reports funding from NIMH (R01MH124839 and R01MH132733). KSO'C reports funding from NIMH (5R01MH124839-02) and the Research Council of Norway (334920). REP reports funding from NIH and NIMH (R01MH125938), and the Brain and Behavior Research Foundation (28632; P&S Fund). HW reports funding from NIMH (R01MH122509) and the National Human Genome Research Institute (NHGRI; UM1HG012003).

JS is funded by a grant from NIMH (MH119746). RS is funded by the Centre for Brain and Mind grant of the Rohini Nilekani Philanthropies. BV is funded by the DBT/Wellcome Trust India Alliance Intermediate Clinical Fellowship (IA/CPHI/20/1/505266). CMN reports funding from NIMH (R01MH106595). EGA is supported by NIH and NHGRI (R01HG012869). The PGC is funded by NIH (MH124871, MH124851, MH124839, MH124847, MH124873, MH124875, and DA054869). All other authors declare no competing interests.

Acknowledgments

The PGC thanks Grace Lutter, Ailiemara Kernan, and NaEshia Ancalade for their expert assistance. Data for figure 2 are frequently updated and can be obtained via email inquiry to the PGC at the following address: pgc.analytics1@gmail.com.

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